EXPLORATION E

Original Paper Autho

Yuri Burda, Harrison Edwards, Amos Stork

Problem/Objective

Something

Meth

Intrinsic Reward

In sparse-reward envi
recieve little to no fee

To encourage explor
component is added t

Total reward become
and intrinsic reward (**i_t**

$$r_t =$$

i_t is large for unfamili
states become familia

Random Networ

1.1

Intrinsic reward is com

BY RANDOM NETWOR

Poster Presentation - Group #8

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ey, Oleg Klimov

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Anthony Nasr, Mariana C

nodology

ironments, RL agents may
dback from environment.

ation, an intrinsic reward
o the reward signal.

s the sum of extrinsic ($\mathbf{e}_t$)
):

$$\mathbf{e}_t + \mathbf{i}_t$$

ar states and decreases as
r.

k Distillation (RND)

1.2

nputed as the MSE

Results

Something

K DISTILLATION

roup Members

havez Flores, Md Mosarraf Hossain, Mohammad

Limitiations

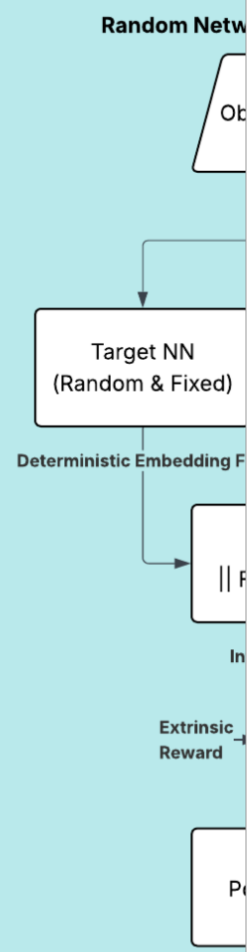
- **Representation Sensitivity:** RND's novelty bonus depends heavily on the random feature space. Poor CNN features can make unrelated states appear similar, causing the agent to miss important distinctions.
- **Reward non-stationarity** : As the predictor improves, the intrinsic reward signal shrinks over time, which can destabilize training if not properly normalized.
- **No environment model** : RND has no understanding of why a state is novel, only that it is. It can get stuck exploring visually complex but strategically irrelevant states.

Why It Matters?

RND directly addresses one of the hardest open problems in deep RL: learning useful behavior when **rewards are extremely** between two neural n

- **Target Network:** R
  fixed.
- **Predictor Network:**
  target.

When encountering a
network produces a
intrinsic reward. As
predictor network
decreases, and the rev



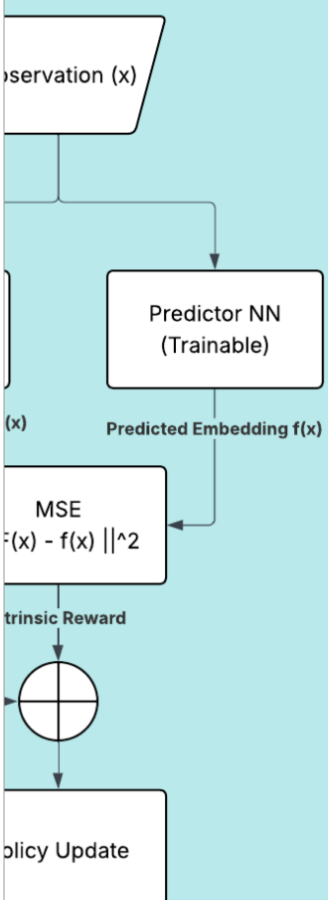
etworks:

andomly initialized &

: Tries to imitate the

new state, the predictor large error → Strong a state is revisited, the learns them, the MSE ward decreases.

ork Distillation Framework



2.2

Key Sources & Acknowl

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rare. It achieves superhuman-level room exploration in Montezuma's Revenge — a benchmark that had stumped RL researchers for years — using a surprisingly simple mechanism. Beyond games, this has real implications for robotics, scientific discovery, and any domain where trial-and-error exploration is costly and feedback is sparse.

Conclusion

RND offers an elegant, scalable solution to the hard exploration problem by repurposing **prediction error** as a **curiosity signal**. By keeping the target network fixed and random, it avoids the noisy-TV pitfall that undermines dynamics-based curiosity methods. The dual reward stream design — combining episodic extrinsic returns with non-episodic intrinsic returns — proves critical for crossing episode boundaries and achieving deep exploration. Its simplicity and compatibility with standard RL algorithms like PPO make it a strong baseline for any sparse-reward setting.

edgements

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